Omar Dennaoui

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Citizenship: Lebanon
Website

Education

University of Louisiana at Lafayette, Lafayette, LA, USA
Ph.D. in Mathematics — Jan 2021 – May 2027 (expected)
Dissertation: *Two Faces of the Segal Condition: Localities of Finite Groups and Cyclic Multicategories* (working title)
Advisors: Justin Lynd and Philip Hackney

University of Balamand, El-Koura, Lebanon
B.S. in Mathematics, with honors — Jan 2017 – Jun 2020

Positions

University of Louisiana at Lafayette
Dissertation Completion Fellow — Aug 2026 – present
Graduate Research Assistant (with Philip Hackney) — Jan 2025 – May 2026
Graduate Teaching Assistant — Jan 2021 – Dec 2024

Research Interests

Finite group theory: fusion systems, partial groups, and localities; the p-local structure of finite groups; p-subgroup complexes and their homotopy type.

Higher category theory: Segal conditions and higher Segal conditions (in the sense of Dyckerhoff–Kapranov); nerve theorems and monads with arities; operads, cyclic operads, and multicategories.

The common thread is the Segal condition: weakening it to measure how far partial groups and localities are from being associative, and transporting it to new operadic settings.

Publications

Contractibility of the orbit space of a saturated fusion system after Steinberg, with Jonathon Villareal.
J. Algebra **677** (2025), 267–277. (doi, arXiv, pdf)

In preparation

Higher Segal conditions and localities of symmetric and alternating groups at the prime 2.

A nerve theorem for cyclic multicategories.

Fellowships & Awards

Dissertation Completion Fellowship (full tuition waiver and stipend), UL Lafayette — 2026 – 2027
Honors Convocation and Academic Excellence Award, UL Lafayette — Spring 2024

Talks

Invited

Topology Seminar, University of Virginia — Oct 2026
Higher Segal conditions and localities of symmetric and alternating groups at the prime 2

Contributed

Representations of Finite Groups and Related Structures, UC Santa Cruz — Aug 2026
Higher Segal conditions and localities of symmetric and alternating groups at the prime 2

International Category Theory Conference (CT 2026), Johns Hopkins University — Jul 2026
A nerve theorem for cyclic multicategories

Southern Regional Algebra Conference, Southeastern Louisiana University — Mar 2026
Higher Segal conditions and localities of symmetric and alternating groups at the prime 2

Southern Regional Algebra Conference & Lloyd Roeling Conference, UL Lafayette — Apr 2025
Contractibility of the orbit space of a saturated fusion system after Steinberg

South Central Topology Conference IV, Texas A&M University — Mar 2025
Contractibility of the orbit space of a saturated fusion system after Steinberg

Zassenhaus Groups and Friends Conference, Texas State University — Jun 2024
Contractibility of the orbit space of a saturated fusion system after Steinberg

South Central Topology Conference III, New Mexico State University — Oct 2023
Contractibility of the orbit space of a saturated fusion system after Steinberg

Local & Expository

Algebra Seminar, UL Lafayette — Oct 2026
Higher Segal conditions and localities of symmetric and alternating groups at the prime 2

Algebra Seminar, UL Lafayette — Nov 2023
Contractibility of the orbit space of a saturated fusion system after Steinberg

Algebra Seminar, UL Lafayette — Mar 2022
Limits and colimits, generators and relations of partial groups

Teaching

Courses taught at UL Lafayette

MTHS 105/105S: Applied College Algebra Fundamentals	Fall 2023, Fall 2024
MTHS 109/109S: Pre-Calculus Algebra	Spring 2024
MATH 105: Applied College Algebra	Spring 2023
MATH 103/104: Applied College Algebra Fundamentals	Fall 2022

Service & Professional Activity

UL Lafayette Graduate Student Chapter of the AMS

President	Fall 2023 – Spring 2024
Secretary	Fall 2022 – Spring 2023
Member	Fall 2021 – present

Memberships

American Mathematical Society	2021 – present
Mathematical Association of America	2021 – present

Computational Skills

Mathematical software: Magma, GAP, Wolfram Mathematica, MATLAB, LaTeX
Programming: Python, C++, Java, JavaScript, SQL
Office: Microsoft Word, Excel, PowerPoint